

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/0__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129__

Does R2 show up as an OSPF neighbor on R1? __Yes__

Does R2 show up as an OSPF neighbor on R3? __No__

What does this information tell you?

____ All traffic from R3 to the 192.168.2.0/24 network (the LAN attached to R2) is being routed ____

____ through R1 instead of going directly to R2 _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

____ R2(config)# router ospf 1 _____

____ R2(config-router)# no passive-interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____ 1 T1 serial link(cost 64) + R2 Gb 0/0 LAN link(cost 1) = 65 _____

Is R2 listed as an OSPF neighbor to R3? __yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____ A higher default reference bandwidth is required for more accurate cost calculation _____
_____ on these faster links _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_____ cost to 192.168.3.0/24: R1 S0/0/1(cost 781) + R3 G0/0(cost 1) = 782 _____

_____ cost to 192.168.23.0/30: R1 S0/0/1 (cost 781) + R3 S0/0/1 (cost 64) = 845 _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____ With each serial link set to a cost of 781, the path to the 192.168.23.0/24 network _____

_____ spans two links, cost = 781 + 781 = 1562 _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_____ OSPF will choose the route with the least accumulated cost. The route with the lowest _____

_____ accumulated cost is 781 + 781 + 1 = 1563 (go through R2) < the cost directly go R3 _____

_____ which is 1565 + 1 = 1566 _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?
_The election of the Designated Router and Backup Designated Router depends on Router ID _____
_ assignments. To avoid instability, the Router ID should not rely on a physical interface that could go down. _
_ Instead, it should be set using a loopback interface address or explicitly with the router-id command ____

2. Why is the DR/BDR election process not a concern in this lab?
_Multi-access networks such as Ethernet and Frame Relay require a DR/BDR election, but point-to-point ____
_ serial links do not. Therefore, this process is not relevant for the links used in this lab _____

3. Why would you want to set an OSPF interface to passive?
_ Configuring a LAN interface as passive eliminates unnecessary OSPF traffic on that interface, freeing up _
_ bandwidth. The router continues to advertise the network to neighboring routers. ____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?
_ OSPF is hierarchical and scalable because of its area-based architecture, where a central Backbone ____
_ (Area 0) connects multiple sub-areas to localize traffic, minimize SPF calculations, and enable route
summarization. _____
5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in
the range 10.0.0.0 to 10.255.255.255?
_ network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF
with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be
used for the routing decision?
_ EIGRP _____
